THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

(KATAVI, MBEYA, NJOMBE, RUKWA, RUVUMA AND SONGWE)

FORM FOUR ZONAL MOCK II EXAMINATION

033/1 BIOLOGY 1

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TIME: 3:00 HOURS **YEAR: 2025**

INSTRUCTIONS

1. This paper consists of section A, B and C with a total of **eleven (11)** questions.
2. Answer **all** questions in section A and B and **two (2)** question from section C.
3. Section A carries **sixteen (16)** marks, section B carries **fifty-four (54)** marks and section C carries **thirty (30)** marks.
4. All writings must be in **blue** or **black** ink, **except** drawing which must be in pencil.
5. All communication devices and any unauthorized materials are **not** allowed in the examination room.
6. Write your **Examination Number** on every page of your answer sheets provided.

SECTION A (16 Marks)

Answer **all** questions in this section.

1. For each of the items (i) – (x) choose the correct answer from the given alternative and write its letter besides the item number in the answer booklet provided.

(i) Kapama is diagnosed with diabetes mellitus. Which control measure would you recommend to Kapama?
A Reduce intake of cholesterol foods
B Limiting amount of water intake
C Reduce intake of sugary foods
D Eating enough starch
E Consuming a high fat food

(ii) Mr. Godwin is working as a nurse employed by the Red Cross. He was called for help in Gaza during the war as many victims had suffered severe blood loss. Which blood group will be suitable for transfusion at that movement?
A AB B A C B
D O+ E O –

(iii) When returning home from school, Helen saw a big scorpion crossing the road. Helen's body suddenly produced a hormone which prepared her for an action. Which hormone and gland were responsible for this?
A Adrenaline from thymus gland
B Aldosterone from adrenal glands
C Adrenaline from adrenal glands
D Antidiuretic hormone from pituitary gland
E Thyroxine from thyroid glands

(iv) What type of movement is shown by a human sperm cell?
A Amoeboid movement B Flagellar movement
C Movement of curvature D Ciliary movement
E Muscular movement

(v) Students at Kazamwendo secondary school were educated about risk behaviors that must be abstained. Which one among the following should be avoided?
A Alcoholism and patience. B Unsafe sexual intercourse and exercising.
C Raping and alcoholism D Sharing books during studying and exercising
E Hardworking

(vi) Which of the following is the correct order of event?
A Ovulation, fertilization, implantation
B Fertilization, ovulation, implantation
C Implantation, fertilization, ovulation
D Fertilization implantation, ovulation
E Ovulation, implantation, fertilization

(vii) Form four student was observing liver fluke by using hand lens. If the actual size of liver fluke was 3cm and the size of the image was 9 cm. What was the magnification power of a hand lens used?
A 27 X B 12X C 3X
D 0.3 X E 2.7X

(viii) Which organ is used to prevent the food from entering the trachea?
A Larynx B Pharynx C Epiglottis
D Tonsils E Sphincters

(ix) Structures with common ancestral origin but with different function
A Vestigial structure B Analogous structure
C Homologous structure D Rudimentary structure
E Speciation

(x) Among the following point out the food conducting cell in plant
A Meristem B Xylem C Phloem
D Epidemies E Guard cell

2. Match the description of Growth and development in **Column A** with the corresponding terminologies of growth in **Column B** by writing a correct letter of the response beside the item number in the item number in the answer sheet provided.

COLUMN A	**COLUMN B**
(i) A type of seed germination where by cotyledons are pushed above the ground (ii) A pore which allows water into a seed during germination (iii) Inability of a seed to germinate (iv) Prevent a viable seed to germinate (v) A type of cell division where by the same number of chromosomes is maintained from parent cell to daughter cell. (vi) Cause the increase in length and height of roots and shoots	A Apical meristem B Lateral meristem C Micropyle D Meiosis E Epigeal germination F Hard seed coat G Mitosis H Dormancy I Hypogeal germination

SECTION B (54 Marks)
Answer **all** questions in this section

3. A group of form four students from Chita secondary school had a study visit to St. Mpindo hospital. They noted a patient with sunken eyes, severe dehydration, weak pulse, deep breathing, vomiting and convulsions.

(a) What could be the possible disease that the patient was suffering from?

(b) Briefly explain the cause and transmission of the disease

(c) Mention preventive measures of the disease mentioned in (a) above (5points)

4. Mr. George prepared a microscopic slide and placed it on the stage of the microscope for student's observation. During observation one student noticed the following parts of a cell; the nucleus, cytoplasm, cell wall, cell membrane, mitochondria, chloroplast and large vacuole.
   (a) What was the name of the cell?
   (b) Name the parts of the cell in which you have used to name the cell in (a) above?
   (c) Identify four specialized plant cells and State their roles.

5. (a) State the importance of respiratory surfaces being;
      (i) Having thin membranes
      (ii) Moist
      (iii) Well supplied with capillaries
   (b) Explain why:
      (i) Oxygen debt is needed after heavy body exercises
      (ii) Is better to breathe through the nose than through the mouth
      (iii) A locust never dies when its head is fully immersed in water

6. (a) In some Tanzanian tribe's people believe that infertility among married couples affects women only. As a form four student disagree by educating them about six (6) factors which can Cause such situation in males
   (b) Give out three (3) significances of the process which take place in somatic cell during growth of an organism

7. (a) State first Mendel´s law of inheritance
   (b) In an experienced varieties of been seeds having red seed coat were crossed with white seed coat beans. All the beans obtained in F_1 generation had red seed coat. When F_1 generation selfed a total of 7324 seeds were obtained
      (i) Use appropriate genetic symbols to work out the genetic ratio of F_2 generation
      (ii) From the information above find the total number of red seeds in F_2
      (iii) Find the percentage of white and red seed coat in F_2

8. (a) Water and salts are needed in constant optimal amount for effective and proper functioning of the body.
      (i) Explain three factors which affect the contents of water and salts in human body.
      (ii) Mention the two hormones which are responsible for regulating the amount of water and in the body.

(b) Explain why form three students at Jordan secondary school have frequently urination during morning class session?

SECTION C (30 Marks)

Answer only **two (2)** questions from this section

9. Irresponsible sexual behavior among youths poses a problem to the Tanzanian community. Justify this statement by elaborating six effects.

10. Body of human being is made up of frame work of interconnected bones which play crucial roles to the body. Justify this statement by using six (6) points.

11. Explain physiological mechanism exhibited by homoiotherms in cold and hot environment